

Project

Airline Data Management and Analysis Using Power BI

Task 1: Data Preparation and Cleaning (10 Marks)

- Extract and transform data in Power Query.
- Clean data: remove duplicates, handle missing values, and format columns.
- Deliverables: Screenshot of Power Query Editor showing cleaned data

Sol:

Objective: Load data and clean it using Power Query.

- Open Power BI > Home tab > Get Data > Excel:
- Load Flight_Information.xlsx, Ticket_Information.xlsx, and Passenger_Information.xlsx.
- Click on "Transform Data" to open Power Query Editor.
- Clean Data:
- Remove duplicates: Right-click a row > Remove duplicates.
- Handle missing values:
- Filter or replace nulls using "Remove Rows" > "Remove Blank Rows" or "Replace Values".
- Format columns:
- Set correct data types (e.g., text for FlightID, numbers for SeatNumber, etc.).

Table with 5 columns: FlightID, FlightNumber, Airline, Destination, Status. The table contains 17 rows of data. The 'Status' column has values like 'On Time', 'Cancelled', 'Delayed', and 'On Time'. The 'Destination' column has values like 'Houston', 'Chicago', 'New York', 'Phoenix', 'Los Angeles', and 'Chicago'. The 'Airline' column has values like 'Airline D', 'Airline B', 'Airline A', 'Airline C', and 'Airline A'. The 'FlightNumber' column has values like 'FL1102', 'FL1435', 'FL1860', 'FL1270', 'FL1106', 'FL1071', 'FL1700', 'FL1020', 'FL1614', 'FL1121', 'FL1466', 'FL1214', 'FL1330', 'FL1458', 'FL1087', and 'FL1372'. The 'FlightID' column has values like '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '11', '12', '13', '14', '15', and '16'. The table is titled 'Table.RemoveColumns(Table.RemoveDuplicates(...))'.

Untitled - Power Query Editor

Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Use First Row as Headers Replace Values Data Type: Whole Number Merge Append Transform

Queries [3]

- flight_information
- ticket_information
- passenger_information

Table.Distinct(*removed columns*, {"TicketID"})

	TicketID	FlightID	BookingStatus
1	Valid	100%	Valid
2	Error	0%	Error
3	Empty	0%	Empty
4	Valid	100%	Valid
5	Error	0%	Error
6	Empty	0%	Empty
7	Valid	100%	Valid
8	Error	0%	Error
9	Empty	0%	Empty
10	Valid	100%	Valid
11	Error	0%	Error
12	Empty	0%	Empty
13	Valid	100%	Valid
14	Error	0%	Error
15	Empty	0%	Empty
16	Valid	100%	Valid
17	Error	0%	Error

Query Settings

PROPERTIES

Name

ticket_information

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Columns
- Removed Duplicates

3 COLUMNS, 50 ROWS Column profiling based on top 1000 rows

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ENG IN 2003 20-05-2025

Untitled - Power Query Editor

Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Use First Row as Headers Replace Values Data Type: Whole Number Merge Append Transform

Queries [3]

- flight_information
- ticket_information
- passenger_information

Table.Distinct(*removed columns*, {"PassengerID"})

	PassengerID	FlightID	SeatNumber
1	Valid	100%	Valid
2	Error	0%	Error
3	Empty	0%	Empty
4	Valid	100%	Valid
5	Error	0%	Error
6	Empty	0%	Empty
7	Valid	100%	Valid
8	Error	0%	Error
9	Empty	0%	Empty
10	Valid	100%	Valid
11	Error	0%	Error
12	Empty	0%	Empty
13	Valid	100%	Valid
14	Error	0%	Error
15	Empty	0%	Empty
16	Valid	100%	Valid
17	Error	0%	Error

Query Settings

PROPERTIES

Name

passenger_information

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Columns
- Removed Duplicates

3 COLUMNS, 100 ROWS Column profiling based on top 1000 rows

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ENG IN 2004 20-05-2025

Task 2: Data Modeling (10 Marks)

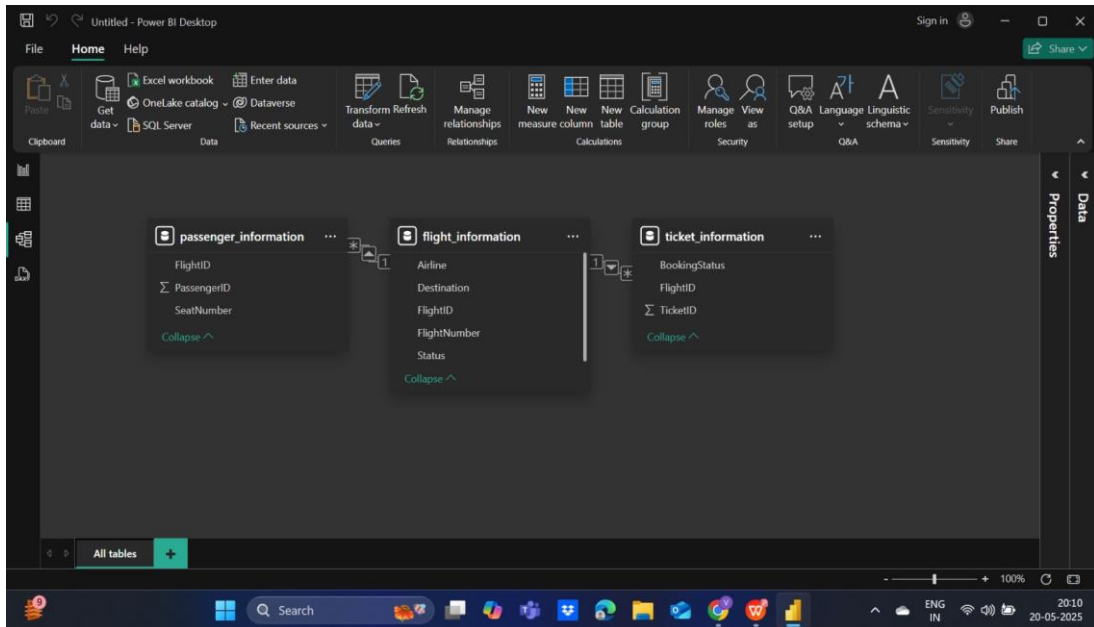
- Create relationships between datasets (FlightID as the key).
- Understand cardinality and configure the model appropriately.
- Deliverables: Screenshot of the data model with relationships.

Sol:

Objective: Create relationships between datasets using FlightID.

- Go to the Model view (third icon on the left).
- Drag FlightID from:
- Flight_Information → to Passenger_Information

- Flight_Information → to Ticket_Information
- Power BI should automatically detect one-to-many relationships.
- Check cardinality:
FlightID in Flight_Information (One side)
FlightID in other tables (Many side)



Task 3: Enhanced Data Insights (10 Marks)

- Add a conditional column to classify flights as "Best" or "To Be Improved" based on status.
- Use "Column from Examples" to extract the flight number from FlightNumber.
- Deliverables: Screenshot of the transformed data.

Sol:

Objective: Add insights with conditional and example-based columns.

- Back in Power Query:
- Add a Conditional Column:
- Name: FlightCategory
- Rule: If Status = "On Time" then "Best" else "To Be Improved".
- Extract Flight Number:
- Use "Column from Examples" to extract the numeric/short code from FlightNumber.

Table with 6 columns: ID, Origin, Destination, Status, FlightStatusCategory, FlightNumberOnly. The table contains 17 rows of flight data.

ID	Origin	Destination	Status	FlightStatusCategory	FlightNumberOnly
1	D	Houston	On Time	Best	1102
2	B	Chicago	On Time	Best	1435
3	A	New York	Cancelled	To Be Improved	1860
4	C	Chicago	Delayed	To Be Improved	1270
5	C	New York	Delayed	To Be Improved	1106
6	A	Phoenix	On Time	Best	1071
7	C	Los Angeles	Cancelled	To Be Improved	1700
8	C	Los Angeles	Delayed	To Be Improved	1020
9	A	Los Angeles	Cancelled	To Be Improved	1614
10	D	Chicago	Cancelled	To Be Improved	1121
11	A	Phoenix	On Time	Best	1466
12	D	New York	Delayed	To Be Improved	1214
13	C	Houston	On Time	Best	1330
14	C	New York	Delayed	To Be Improved	1458
15	C	Houston	Delayed	To Be Improved	1087
16	B	New York	Delayed	To Be Improved	1372
17					

Task 4: Calculations Using DAX (10 Marks)

● Calculate:

- Total passengers for a specific flight.
- Total tickets booked.
- Filtered table showing "Best" flights only.

● Deliverables: Screenshot of DAX calculations and results.

Sol:

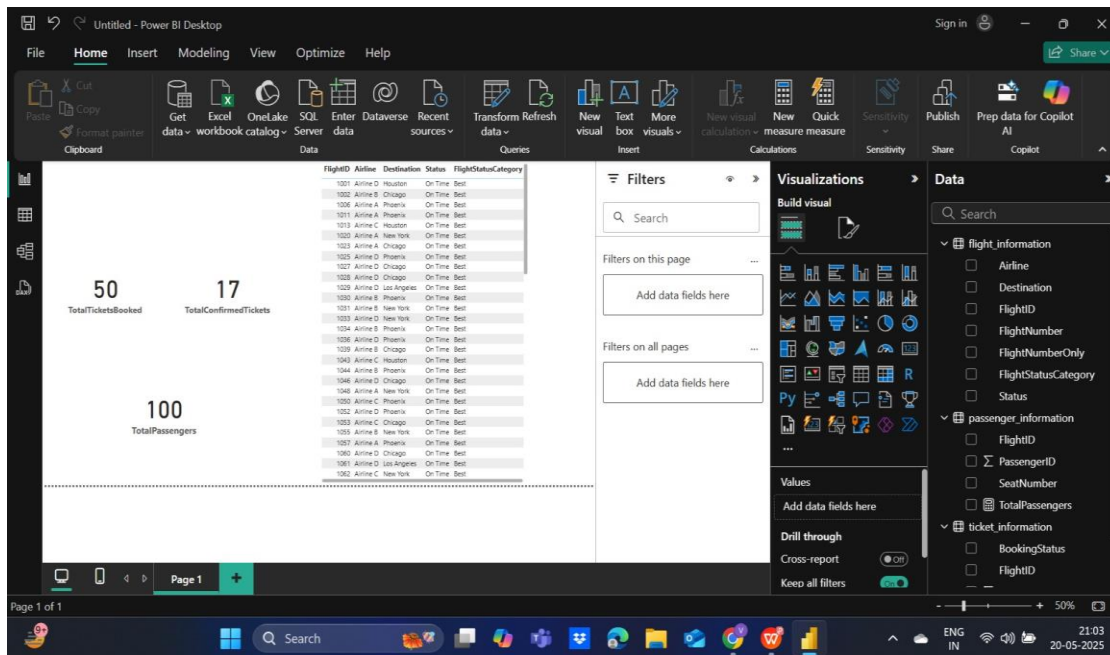
Objective: Write DAX for analytical insights.

Go to Modeling tab → Click New Measure.

Enter the following formula:

DAX Formula's :

- **TotalPassengers = COUNT(Passenger_Information[PassengerID])**
- **TotalTicketsBooked = COUNTROWS(Ticket_Information)**
- **TotalConfirmedTickets =**
- **CALCULATE(COUNTROWS(Ticket_Information), Ticket_Information[BookingStatus] = "Booked").**



Task 5: Visualization and Interactive Features (20 Marks)

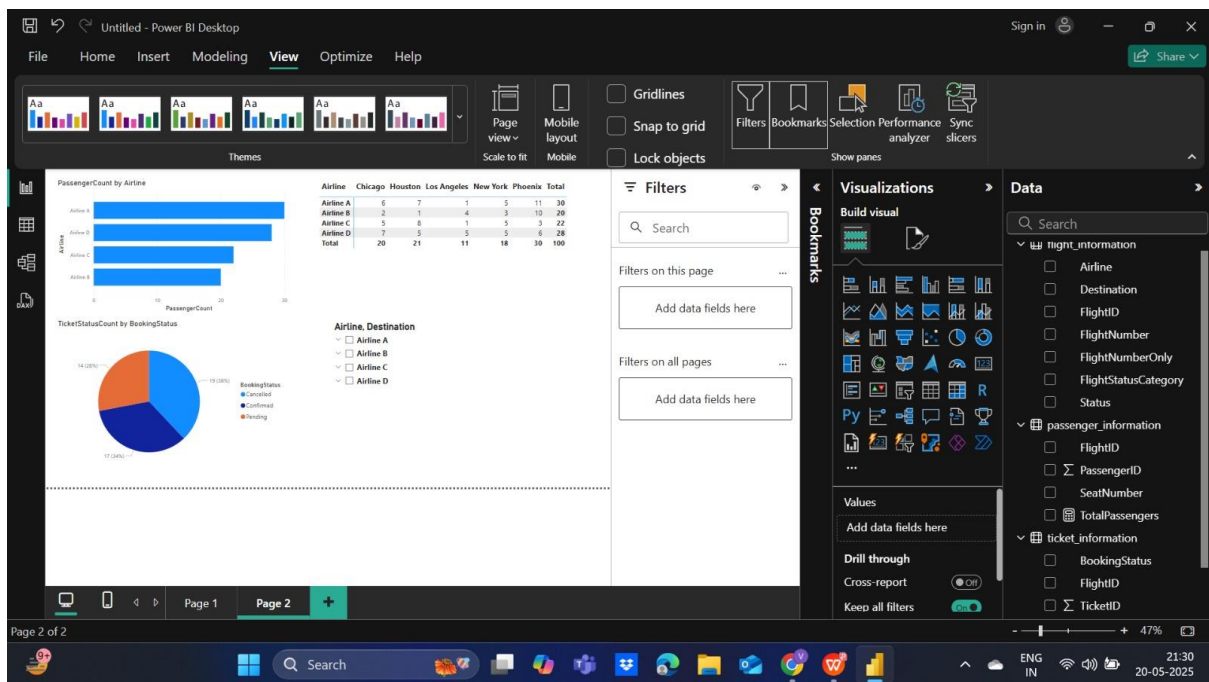
- Create visuals for:
 - Passenger count by airline.
 - Ticket booking statuses.
 - Flights by airline and destination.
- Add interactive features for:
 - Destination and Airline.
 - Quick views.
 - Airline-specific pages.
- Deliverables: Screenshots of all visuals and interactive features.

Sol:

Objective: Create Visuals

- Visuals to Create:
 - Passenger Count by Airline:
 - Use bar chart: Axis = Airline, Value = TotalPassengers.
 - Ticket Booking Status:
 - Use Pie chart: Legend = BookingStatus, Values = Count of TicketID.
 - Flights by Airline and Destination:

- Use clustered bar chart or map visualization.
- Interactivity:
- Add slicers:
- Destination
- Airline
- Quick views



Task 6:

Final Dashboard and Power BI Service (20 Marks)

- Design a comprehensive dashboard with key visuals and insights.
- Configure Row-Level Security (RLS) for Airline A data and assign it to a user.
- Set up a schedule refresh at 5 PM daily.
- Deliverables: Screenshot of the published dashboard and RLS configuration.

Sol:

Design the Dashboard:

- Arrange key visuals, slicers, and KPIs.
- Use titles, colors, and layout for clarity.

Configure Row-Level Security (RLS):

- Model view > Manage roles:

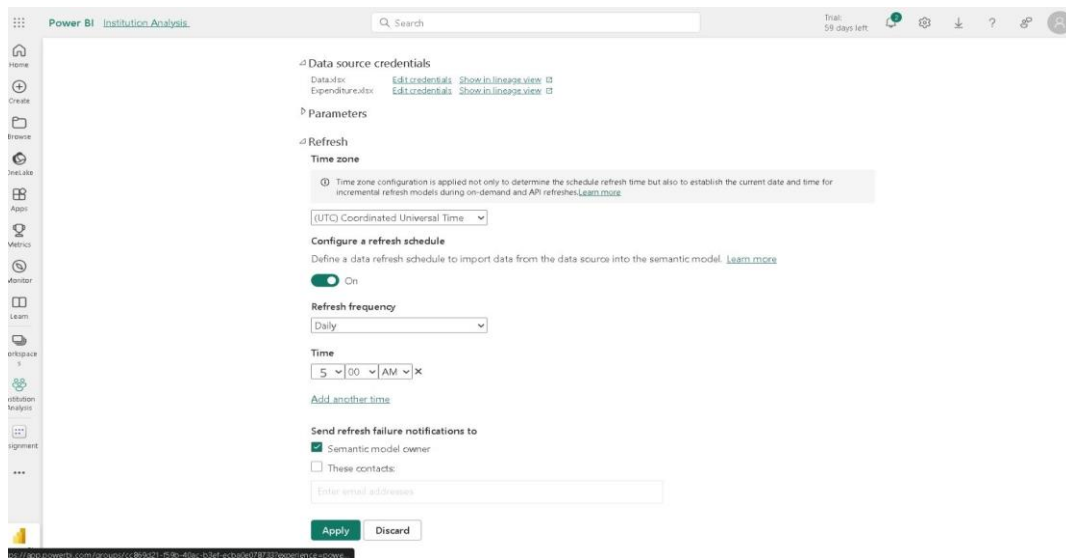
- Create Role: Airline_A_Access
- Table: Flight_Information
- Rule: [Airline] = "Airline A"
- Assign user:

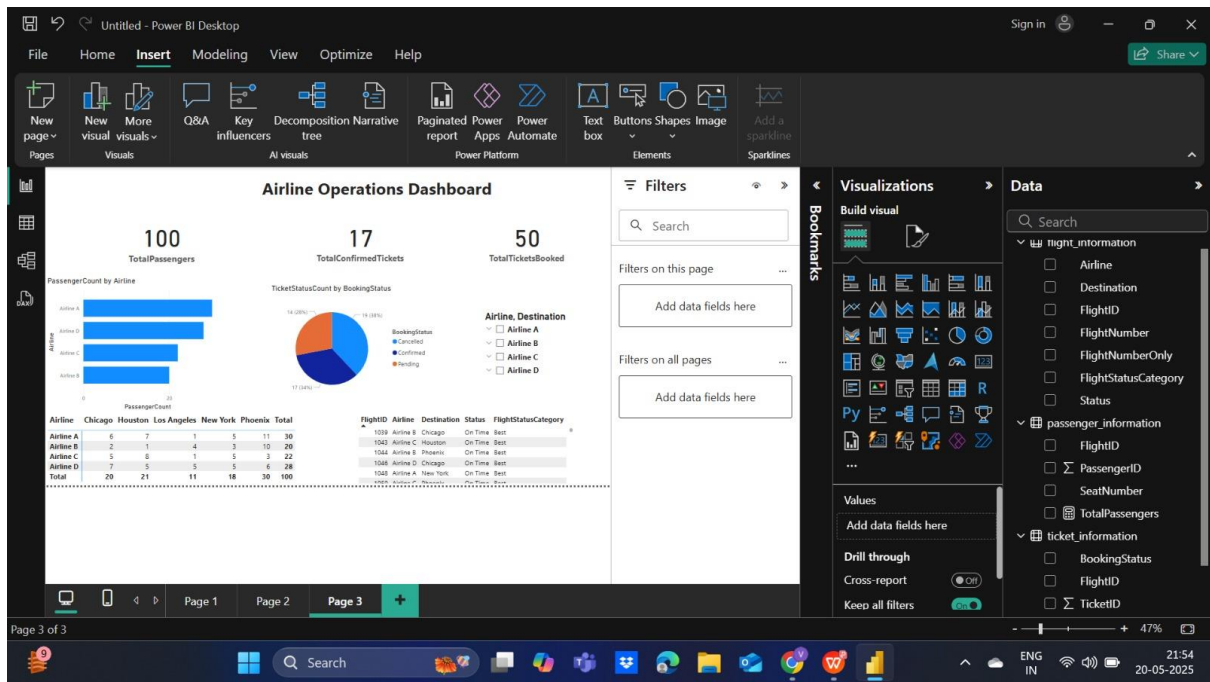
Publish to Power BI Service.

- Go to the dataset → Security → Add users to the Airline_A_Access role.

Set Scheduled Refresh (Daily at 5 PM):

- Publish your report to Power BI Service.
- Go to Workspaces > Dataset > Settings.
- Under Scheduled Refresh:
- Turn on refresh.
- Set time to 5:00 PM.
- Save settings.





Project Submission Link:

<https://drive.google.com/file/d/1g6YxU22oGMSOj7Wf91i6YFsnjNmbtEk3/view?usp=sharing>